

SPIDER-MAN: 3D OPEN- WORLD ACTION- ADVENTURE

The Ultimate Game Prompt (PC · offline-first · optional online)

Version 1.0 · Researched 28 Sep 2026 ·
Primary engine: Unreal Engine 5.8

How to use this file (for you, not the AI)

1. Fill in **Section 0 (MY SETUP)** in Part A, especially your PC specs.
2. Paste **all of Part A** into your AI assistant as the first message.
3. Send the **Part B phase prompts** one at a time. Don't advance until the phase's

Definition of Done is met.

4. Be realistic: no single prompt outputs a finished AAA game. Final art, animation, voice acting and music are big production jobs. This prompt gives an AI (or a human team) a complete blueprint and a build order that reaches a polished, playable slice first, then grows into the full game.
5. IP note: Spider-Man and related characters belong to Marvel/Sony. That's fine for private learning and fan projects. If you ever want to release or sell the game, swap in an original hero (see Section 19).

PART A: MASTER PROMPT

(Copy from here to the end of Part A.)

0. MY SETUP (fill in; if blank, assume sensibly and say what you assumed)

- PC: CPU ____ | GPU ____ (VRAM ____ GB) | RAM ____ GB | Free SSD space ____ GB | OS ____ | Monitor ____ (resolution / Hz)
- My skill level: [never coded / beginner / intermediate / professional]
- Team: [solo / ____ people] · Hours per week: ____ · Target date for a playable slice: ____
- Engine: [Unreal Engine 5.8 (default) / Unity 6.3 LTS / Godot 4 / you choose based on my PC]
- Input: [keyboard + mouse / gamepad / both]
- Online: [offline only / offline + optional 2-player co-op / plus leaderboards]
- Asset budget: [free / AI-assisted assets only / \$____]

- Scope now: [vertical slice first (recommended) / full game]

1. YOUR ROLE AND RULES

You are a full game studio in one: game director, lead designer, principal gameplay and engine programmer, technical artist, animator, audio director, QA lead and producer, with AAA shipping experience on open-world action-adventure games for PC. Help me design and build the game below, step by step, to a professional standard.

Rules of engagement:

- Work in the phases of Section 20. Never jump to polish before the core feel is proven.
- Give complete, compilable, commented code and exact editor steps (menu paths, settings, asset names). No pseudo-code for core systems and no "TODO: implement".

- Be honest about limits. If something cannot be produced as text (final art, animation, motion capture, voice, music), say so, then give a concrete asset plan (source, format, cost, import steps).
- Prefer the simplest approach that hits the quality bar. Explain trade-offs in 1 to 3 lines.
- Keep every tunable number in data assets or config files so I can adjust feel without recompiling.
- After each deliverable, list what to test and what "good" looks like.
- Ask at most 3 clarifying questions, and only if truly blocking. Otherwise state your assumption and proceed.

2. HIGH CONCEPT

- **Working title:** SPIDER-MAN: WEB OF THE CITY (placeholder)
- **One-liner:** Swing through a dense, living New York, unravel a conspiracy built on

the city's own power grid, and balance Peter Parker's life with Spider-Man's.

- **Genre:** 3D open-world action-adventure, third-person.
- **Modes:** Single-player campaign, fully playable offline. Optional online (co-op, leaderboards, cloud saves) that never gates the campaign.
- **Platform:** Windows 10/11 64-bit, PC first (Steam-ready). Keyboard + mouse and gamepad.
- **Rating target:** ESRB T / PEGI 12. Action violence, no gore.
- **Length:** Main story 12 to 15 hours, 30 to 40 hours with side content, 60+ hours for completionists.
- **First hour must include:** a swing within the first minute, a first fight within 5 minutes, a first boss or big set-piece by about 45 to 60 minutes.
- **Tone:** Heartfelt, witty, cinematic. Superhero fantasy with grounded human stakes.

3. DESIGN PILLARS (every decision must serve these)

1. **Traversal is the game.** A first-time player must feel like Spider-Man in their first swing. Fun and flow beat physical accuracy. (Lesson from swing games: pure physics is too hard and knocks players out of flow, so stylize and assist.)
2. **Fluid, readable, acrobatic combat.** Improvise with dodges, web-strikes, gadgets and the environment. Never a chore, always readable.
3. **A living, reactive city.** Crowds, traffic, events, weather and radio commentary. Both rooftops and streets feel alive.
4. **Heart.** Peter's personal life and Spider-Man's duty collide, and the story earns its emotion.
5. **Smooth and responsive.** 60 FPS baseline, low input latency, no hitches. When feel and fidelity conflict, feel wins.

6. **Respect the player.** Deep accessibility, generous options, no forced online, no paywalled gameplay, fair difficulty.

4. TECH STACK, HARDWARE TARGETS, PERFORMANCE BUDGETS

4.1 Engine and tools (current as of Sep 2026)

- **Primary: Unreal Engine 5.8.** Released June 2026 and the last planned major UE5 release; UE6 early access is targeted for the end of 2027. Start on 5.8 and do not plan a mid-project migration.
- **Use these UE features:** Nanite; Lumen (including the lighter "Lumen Lite / medium" mode for mid-range GPUs); MegaLights (production-ready in 5.8); Virtual Shadow Maps; World Partition + HLOD + Data Layers; PCG framework (procedural city dressing); Chaos

(physics, destruction, cloth); Niagara (VFX); Motion Matching (Pose Search); Control Rig; Motion Warping; Gameplay Ability System (abilities, gadgets, combat); Enhanced Input + Common UI; StateTree / Behavior Trees / EQS (AI); Mass Entity + ZoneGraph for crowds and traffic (verify maturity in 5.8); MetaSounds (adaptive audio); Sequencer (cutscenes); Iris replication (production-ready in 5.8) for co-op; Steam / EOS Online Subsystem.

- **Movement note:** Mover was improved in 5.8 but is still moving out of Experimental status. Default to a custom C++ movement component for traversal (predictable and fully controllable) and evaluate Mover on a side branch.
- **Languages:** C++ for core systems (movement, combat, AI, networking, save). Blueprints for content, UI and prototyping. Python / Editor Utility Widgets for tools.

- **AI in the editor (optional):** UE 5.8 ships an experimental MCP plugin that lets AI models operate inside the editor. Use it if available to speed up level and asset setup.
- **Alternatives, only if my hardware or skills demand it:** Unity 6.3 LTS (HDRP, C#, supported through Dec 2027; needs more work to reach AAA-level realism) or Godot 4 (light, great for prototypes, not for AAA realism). If you recommend an alternative, justify it from my PC specs and skill level, then adapt every section.
- **Pipeline:** Git + Git LFS (or Perforce) from day one; nightly builds; coding standard; naming conventions; asset budgets tracked in a spreadsheet.

4.2 Hardware tiers (design to these; anchored on real data)

Reference points: Marvel's Spider-Man 2 on PC lists a GTX 1650 / RX 5500 XT class GPU as minimum (720p, 30 FPS, very low) and an

RTX 3060 / RX 5700 class GPU as recommended (1080p, 60 FPS, medium), with 16 GB RAM and a mandatory SSD. The RTX 3060 was also the most common GPU on Steam in July 2026, and 16 GB of RAM was the most common memory size.

- **Minimum:** GTX 1650 / RX 5500 XT (4 GB VRAM) · 4-core CPU (i3-8100 / Ryzen 3 3100 class) · 16 GB RAM · SSD · 1080p upscaled → 30 FPS · Low preset, no ray tracing, lightweight GI. (UE5 open worlds are CPU-heavy, so verify on a real 4-core.)
- **Baseline (must be smooth):** RTX 3060 12 GB / RX 6600 XT · 6-core CPU (Ryzen 5 3600 / i5-12400 class) · 16 GB RAM · SSD · 1080p → **60 FPS** with upscaler on Quality · High preset, Lumen Lite.
- **High:** RTX 4070 / 5070 / RX 9070 · 8-core CPU · 32 GB RAM · 1440p → 60 to 120 FPS · Epic preset, optional hardware ray-traced reflections.
- **Ultra:** RTX 5080+ / RX 9070 XT+ (16 GB+ VRAM) · 8-core X3D-class CPU · 32 GB

RAM · 4K → 60+ FPS with upscaling ·
ultrawide 21:9 to 32:9 · full ray-tracing
options.

Task: classify MY PC (Section 0) into one of
these tiers, set the default scope and
graphics preset accordingly, and tell me
plainly what to expect.

4.3 Performance budgets (hard gates)

- 60 FPS baseline: frame time ≤ 16.6 ms
(GPU ≤ 14 ms, game thread ≤ 8 ms, render
thread ≤ 8 ms). 1% lows ≥ 45 FPS. **No
hitch longer than 50 ms while swinging at
top speed.**
- Input-to-screen latency $\leq$ about 80 ms on
the Baseline tier.
- Memory at Baseline (1080p, High): VRAM
 ≤ 8 GB, RAM ≤ 12 GB. Install size ≤ 100
GB.
- Loading: cold boot to gameplay ≤ 25 s
from SSD; fast travel ≤ 5 s; respawn ≤ 3 s.

- Streaming: no visible pop-in at 60 m/s swing speed. Use predictive streaming along the velocity vector.
- A milestone is NOT done if it regresses these numbers.

4.4 Rendering strategy

Nanite for static geometry. Lumen (Lite on Baseline, hardware RT on High/Ultra).

MegaLights for dense night lighting. Virtual Shadow Maps. DLSS, FSR and XeSS upscaling, plus optional frame generation (never required to hit Baseline). Dynamic resolution scaling. PSO precaching plus a first-launch shader warm-up to remove stutter. HDR10. Ultrawide up to 32:9 with correct cinematics. Scalability presets with per-setting toggles and a built-in benchmark.

5. STORY, CHARACTERS AND TONE

(Original story. Classic Marvel characters are used as swappable placeholders.)

Premise: Five years into being Spider-Man, Peter Parker is stretched thin: a struggling researcher, behind on rent, worried about Aunt May. Then Aegis Dynamics launches the "Aegis Grid", a city-wide, AI-managed power and security network championed by its charismatic founder Elena Voss (original character). Blackouts and "accidents" follow, and old rogues emerge, each tied to the Grid. Peter must find out who is really pulling the strings without losing the people he loves.

Cast

- Peter Parker / Spider-Man: witty, exhausted, principled.
- Aunt May: retired nurse who volunteers at a neighborhood clinic; his moral compass.
- Mary Jane Watson: investigative journalist; short playable

stealth/investigation interludes (max 5% of playtime).

- Harry Osborn: childhood friend who works at Aegis; torn loyalties.
- Captain George Stacy: NYPD ally.
- Elena Voss: Aegis CEO; sincere, dangerous idealist.
- Villains: Vulture (aerial scavenger-tech), Electro (Grid technician, overloaded), Sandman (brute force, big destruction), Mysterio (illusion/AR designer at Aegis), Doctor Octopus (Aegis chief engineer; final boss alongside the Grid AI).

Themes: responsibility, trusting systems versus trusting people, community.

Structure: 3 acts, 16 story missions (Section 8), about 8 full cinematics plus in-engine scenes, about 3 hours of dialogue. Produce a writers' bible: voice rules per character, banter style for combat and swinging, continuity tracker.

6. CORE GAMEPLAY SYSTEMS

6.1 Traversal (most important system; build and tune it first)

States: Ground (walk/run/sprint), Air, Swing, Zip, Launch, Glide (web-wings, unlocked mid-game), Wall-Run, Wall-Cling/Crawl, Ledge-Grab/Vault, Perch, Dive/Free-fall, Stagger/Ragdoll, Interact. Use a clean hierarchical state machine with explicit transitions.

Swing model (hybrid: physics-flavored, assist-driven)

- Hold the swing input to fire a web. The system auto-selects an anchor using a sphere-trace cone along the camera-forward direction (default 90-degree cone, 80 m reach, prefers points 35 to 55 degrees above the horizon on buildings, lampposts, cranes, billboards, bridges). Show a subtle anchor preview.

- The rope is a variable-length constraint. Motion is a damped pendulum plus designer assists: gravity multiplier 1.5 to 2.5x real, tangential speed boost through the low point of the arc, a minimum-speed floor so you never stall mid-swing, and forgiving anchor re-selection.
- Releasing near the low point of the arc gives a "release boost" (timed release for a bonus). Chained swings build speed tiers (starting values: about 25 → 45 → 70 m/s), with camera FOV, wind audio and motion blur scaling with speed.
- **Never dead-end.** In wide-open spots (parks, avenues, water) offer fallbacks: web-zip to a distant anchor, point-launch, glide, or a limited "sky anchor" (tunable).
- **Anti-frustration:** coyote time, input buffering, auto web-catch before street impact (toggle in accessibility), soft magnetism toward valid anchors, no instant fail when brushing walls (auto wall-run or vault).

Other moves: Web-Zip (fast point-to-point pull, 0.3 to 0.5 s); Point-Launch (springboard to ledges); Wall-Run up full skyscrapers; Air tricks (style inputs grant small speed and XP); Dive-bomb; Superhero landing; Parkour assist (auto vault/mantle); Perch/crouch; Glide (strong forward speed, bank turns, seamless hand-off to and from swinging).

Fast travel: subway stations (≤ 5 s, loading masked) plus an optional stylish skip.

Traversal content: races, ring/gate challenges, time trials, vertical routes with collectibles, ghost replays and leaderboards.

Tuning: everything above lives in a "TraversalTuning" data asset (gravity scale, rope min/max length, anchor cone and reach, spring/damping, boost values, speed tiers, air control, FOV curve, camera lag). Provide a live-tuning debug panel plus recorded test routes (for example, swing the same 2 km route and log time and minimum speed).

Feel gate: in a greybox city, a new tester can swing 2 km without touching the ground after under 5 minutes of practice and rates the feel at 8/10 or better.

6.2 Combat

- **Core loop:** light attack chain; heavy/launcher; dodge (Spider-Sense triggered); perfect dodge (slow-motion window); parry/counter with readable cues; web-strike (yank an enemy to you, or launch yourself at an enemy); aerial combat (juggles and air finishers); environmental takedowns (web to wall, throw props, use vehicles); gadget use mid-combo; Focus meter that fuels cinematic finishers.
- **Readability:** color-coded telegraphs (blockable, unblockable, ranged), directional Spider-Sense indicators, audio cues. A 1-versus-10 fight must stay readable.

- **Attack tokens:** cap simultaneous attackers (2 easy / 3 normal / 4 hard) so fights are fair and cinematic.
- **Game feel ("juice"):** input buffering; animation-cancel windows; hit-stop of 40 to 80 ms on heavy hits; subtle camera shake and FOV kick; layered impact SFX; spark/dust VFX; controller haptics; physics-blended enemy hit reactions; slow motion on finishers only.
- **Enemy roster:** Thug (melee), Shield (blocks; break with heavy or gadget), Brute (unblockable slams), Gunner (ranged), Sniper (laser sight), Drone (flying), Turret/Mech (armored), Tech-Suit (electric), Elite (mixed moveset), Mini-bosses, Bosses.
- **Bosses:** 3 phases, clear telegraphs, traversal-integrated (chases across rooftops, fights in the air), unique arenas, checkpoint before and between phases, cinematic intro and outro, "learn, test, twist" structure.

- **Difficulty:** per-axis sliders (damage, enemy aggression, timing windows, stealth detection, traversal assist) plus 4 presets.

6.3 Stealth

Perch and ceiling-hang; silent takedowns (ledge, ceiling, web-pull, ground); distractions (web-grenade, thrown objects); enemy awareness states (Unaware → Suspicious → Alert → Searching) with visible and audio indicators; sight cones plus hearing radius; security cameras and drones; non-lethal only. Always leave an escape route: never force a restart on stealth failure, degrade gracefully into combat.

6.4 Spider-Sense and Gadgets

- **Spider-Sense:** passive threat ripple plus audio; triggers dodge windows; optional Sense-Vision highlights clues and interactables (investigations, puzzles).

- **Gadgets (6, upgradable):** Impact Web (pin enemy or object to surfaces); Web-Grenade (area entangle); Trip-Mine; Electro-Web (shock, synergy with water); Spider-Drone (distraction or attack); Web-Line Pull (rip away shields and objects). Each has cooldown or ammo, 3 upgrade levels and combo synergies. Gadget wheel with slow motion.

6.5 Progression, Suits, Economy

- XP from actions (style-weighted), levels 1 to 50.
- **4 skill trees** (Traversal, Combat, Stealth, Tech), 12 to 15 nodes each; free respec.
- **Suits:** 12 unlockable (classic plus original designs), each with a unique passive perk and 3 mod slots, earned through challenges; cosmetic recolors.
- **Currencies:** 4 activity-based tokens (Street, Lab, Landmark, Challenge), earned in the world. No microtransactions.

- **New Game+:** carries over suits and skills, harder enemies, extra suit mods.
- **Pacing:** a new ability, gadget or suit every 30 to 45 minutes of play.

7. OPEN WORLD AND LIVING CITY

- **City:** dense, vertical, Manhattan-inspired plus outer-borough areas; about 8 to 12 km² playable; building heights 20 to 300 m. **Rule: no traversal dead zones.** Every street canyon has viable anchors within reach, and rooftop props (water tanks, antennas, cranes) double as anchor points.
- **Districts** (each needs an identity, verticality profile, enemy faction, landmarks and traversal flavor):
 1. Midtown: glass skyscrapers, best swinging, Aegis Tower.
 2. Financial District: narrow canyons, precision swinging, old towers.

3. Chinatown / Lower East: dense low-rise, markets, fire escapes; stealth and street fights.
 4. Harlem-style neighborhoods: brownstones, block parties, community events.
 5. Central Park-style park: open, few anchors; glide and tree-web; calm pacing.
 6. Harbor and Industrial: cranes, containers; heavy enemies; big set-pieces.
 7. Queens-style suburbs: low houses; Peter's home; wall-run over rooftops.
 8. Underground: subway (fast travel) and tunnels (stealth).
- **World tech:** World Partition streaming with predictive loading along the velocity vector; Nanite buildings; HLOD and imposters; PCG for rooftop clutter, facades and street furniture, combined

with hand-authored landmarks and hero blocks; interiors as level instances.

- **Simulation:** 24-hour day/night cycle (about 48 real minutes), weather (clear, overcast, rain, storm, fog, a snow event), wet-surface response, dynamic reflections on glass.
- **Life:** Mass-style crowds (LOD-driven), traffic on lane graphs, ambient events (street performers, protests, block parties), citizens who react (film, cheer, flee), and news/radio/podcast commentary that responds to what the player did (missions, style, collateral damage).
- **Dynamic event director:** spawns crimes and events by time, district, player state and cooldowns; never interrupts a mission; keeps event density tuned.

8. MISSIONS AND ACTIVITIES

8.1 Mission template (use for every mission)

Intro (cinematic or in-world) → objective chain (3 to 7 steps) → checkpoints every 60 to 90 s → clear, fair fail states with quick retry → rewards → outro. Write a one-page doc per mission: goal, "feel target", mechanics used or introduced, enemy set, estimated duration (10 to 25 min), scripting needs, test cases.

8.2 Story missions (16)

Act I: Blackout

1. Rooftop Patrol (tutorial): first swing in under 60 s; wall-crawl; first fight.
2. Aegis Unveiled (chase): tail an armored truck; web-zip, point-launch; Impact Web.
3. Night Shift (stealth): warehouse infiltration; perch, takedowns.
4. Wings Over Midtown (boss: Vulture): aerial combat; zip-to-enemy.

5. The Lab Breach (investigation/puzzle):
Sense-Vision clues; circuit puzzle.

Act II: Short Circuit

6. Live Wire (Electro reveal): power-station set-piece; environmental puzzles.
7. Grid Down (city blackout): night mode; rescues; crowd panic.
8. Sandstorm (mini-boss: Sandman): construction site; large-scale destruction.
9. Human Cost (MJ interlude, playable): stealth plus investigation; emotional beat.
10. Smoke and Mirrors (Mysterio): illusion set-piece; reality-shifting spectacle.
11. Broken Web (low point): stripped-down abilities; earn them back.

Act III: Reconnection

12. Signal Fire: allies rally; AI-ally combat.
13. Ascend: vertical assault up Aegis Tower; wall-run plus fighting.
14. Eight Arms (boss: Doctor Octopus, phases 1 and 2): arena fight.
15. Aegis Zero (final boss: Doctor Octopus plus Grid AI drones): multi-phase; uses every

ability.

16. Home (epilogue): emotional close;
unlocks New Game+ and post-game.

8.3 Open-world activities (each with 3+ variants and scaling rewards)

- Crimes in progress (robbery, car chase, hostage, fire/rescue, drone attack, gang fight)
- Enemy hideouts (stealth or combat)
- Aegis surveillance towers (climb/puzzle to unlock map regions)
- Lab challenges (circuit routing, physics, spectrum puzzles)
- Traversal challenges (races, gates, time trials, glide runs)
- Collectibles (landmark photos, lore audio logs, hidden Easter eggs)
- Citizen requests (mini-stories: deliver by swinging, rescue, help a vendor)
- Optional rogue encounters (villain side arcs)

- Photo-mode challenges
- Community events

Rules: no filler. Every activity must teach or test a mechanic and give a meaningful reward, and the "one more thing" loop should take under 5 minutes.

9. GRAPHICS AND ART DIRECTION (very realistic, cinematic)

- **Look:** cinematic hyper-real, like a big-budget superhero film: reflective glass towers, wet night streets, golden-hour swings, saturated suit colors, speed streaks and wind particles at high speed.
- **Lighting and rendering:** Lumen GI and reflections (Lite on Baseline, hardware RT on higher tiers); MegaLights for dense night scenes; Virtual Shadow Maps; volumetric clouds, fog and god-rays; contact shadows; physically-based

materials; subsurface skin; strand-based hair; Chaos cloth for suit, jackets and flags; wetness and puddles; HDR10; toggles for film grain, chromatic aberration, motion blur and depth of field; ultrawide support.

- **Hero suit:** high-detail fabric normal maps, tension/wrinkle maps, lens deformation on the mask, dynamic dirt and damage, 12+ suit variants.
- **Characters:** MetaHuman-based or custom facial rigs; facial capture for main cutscenes; NPC variety via MetaHuman Collections (experimental in 5.8) or modular kits; LOD chains.
- **Environments:** modular facade kits plus PCG dressing; photogrammetry-based PBR materials; decals, signage, billboards; enterable interiors; Chaos destruction for windows, cars and props.
- **VFX (Niagara):** web strands (ribbon plus rope sim), impacts, sparks, debris, dust,

fire, water splashes, electricity, speed lines.

- **Starting budgets (adjust after profiling):**
hero 120 to 150k triangles; main NPC 40 to 60k; crowd NPC 10 to 20k; 4K textures for the hero, 2K for NPCs; Nanite for buildings and props.

10. ANIMATION AND PHYSICS

- **Animation:** Motion Matching for locomotion; layered and additive animation for combat; procedural swing pose blending (body aligns to the rope tangent, limbs react to speed); foot and hand IK on surfaces; contextual transitions (swing → wall-run → vault → glide); Motion Warping for attack alignment; root-motion combat with cancel windows; additive hit reactions; ragdoll with animation blend-out; cloth and hair simulation.
- **Physics:** web rope constraint (deterministic where networking needs it);

Chaos destruction with strict budgets (glass, stone, cars); arcade vehicle physics; ragdoll limits for performance.

- **Animation source plan:** list every clip needed, then give a sourcing plan (mocap libraries, retargeting workflows, marketplace packs, in-house keyframing) and the retarget steps.

11. CAMERA AND CONTROLS

- **Camera:** third-person spring arm; speed-scaled FOV (base about 80 degrees, rising to roughly 100 to 110 at top speed; player-adjustable 70 to 120); look-ahead during swings; smart collision; lock-on framing in combat; cinematic camera for finishers; free-cam photo mode.
- **Controls:** fully rebindable, with hold/toggle options, sensitivity curves, dead-zones and invert options. Input buffering everywhere. Sensible aim assist.

- **Default gamepad layout (example):** LS move · RS camera · R2/RT hold = swing · A jump / zip · X light attack · Y heavy attack · B web-strike · R1/RB dodge · L1/LB gadget wheel · L2/LT aim gadget · R3 lock-on · L3 sprint / wall-run.
- **Keyboard + mouse:** propose a complete default table (WASD, mouse look, a held key or button for swing) and get my approval before locking it in.
- Haptics and adaptive triggers where supported, with an on/off slider.

12. UI / UX

Minimal HUD (health, Focus, gadget cooldowns, objective marker, mini-map or compass), with every element toggleable. World-space markers. Mission log. Full map with fog-of-war, filters and fast travel. Skill-tree and suit screens with 3D preview. Gadget wheel. Photo-mode UI. Pause and settings. Onboarding hints that fade away. Optional damage numbers. Localization-

ready string tables. UI scale 50 to 200%. Full navigation parity across keyboard, mouse and gamepad. Consistent UI sounds. Loading screens with short tips only.

13. AUDIO AND MUSIC

- Adaptive orchestral-hybrid score: layers respond to speed, combat and danger; stingers on finishers; calm layers while exploring.
- Spatial city ambience: traffic, crowds, wind that rises with height and speed, Doppler on vehicles.
- Layered SFX for webs, impacts and suit movement.
- Reactive dialogue: combat banter, swing quips, radio/podcast commentary.
- VO plan: temporary text-to-speech first, then professional VO; lip-sync pipeline.
- Dynamic mix with ducking, audio occlusion, voice limiting.

- Settings: master, music, SFX, voice, ambience. Subtitles with speaker labels, size and background options.

14. AI SYSTEMS

- **Enemies:** Behavior Tree or StateTree plus EQS; perception (sight and hearing); squad coordination through attack tokens; flanking; cover use for gunners; shared alert states; difficulty scaling; boss phase state machines.
- **Civilians:** flee, film, cheer; avoid the player; navmesh pathing; LOD'd behavior.
- **Traffic:** lane-graph agents with yield and avoid logic; react to hazards.
- **Event director:** see Section 7.
- **Allies:** companion AI for story missions (also the fallback for co-op).
- **Performance:** AI LOD (distant agents tick rarely), time-sliced perception, pooling.
Starting caps: about 12 active combat AI,

about 300 nearby crowd agents,
thousands of cheap far agents.

15. ONLINE AND OFFLINE

Rule #1: the entire campaign and all core content run 100% offline. No always-online requirement and no account needed. Online features are optional and degrade gracefully (hidden or disabled when offline).

Optional online (build only after the offline vertical slice works)

- 2-player co-op (up to 4 later) in free-roam and selected missions; drop-in/drop-out; friends-only invites through Steam/EOS relay; shared world with personal progression saved locally.
- **Netcode:** Iris replication; listen-server host; client-side prediction for movement. For swinging, replicate compact state (anchor point, rope length, swing phase, input) and let each client simulate rope visuals locally with smooth reconciliation.

Crowds and traffic are cosmetic and non-replicated (deterministic seed). Enemies and mission state are replicated. Latency target: playable to about 100 ms, graceful to about 150 ms. Document the bandwidth budget.

- Leaderboards and ghosts for time trials (validated through recorded input/replay); optional cloud saves; photo sharing.
- Anti-cheat: co-op is friends-only; validate leaderboard entries server-side; nothing pay-to-win.

16. SAVE, SETTINGS, PC INTEGRATION

- **Saves:** auto-save at checkpoints plus manual slots with thumbnails; save-version migration; optional cloud saves; corrupted-save recovery; "Continue" reachable from the main menu in 2 clicks or fewer.

- **Settings:** display mode, resolution, ultrawide, refresh rate / VSync / frame cap, upscaler (DLSS / FSR / XeSS) and quality, frame generation on/off, dynamic resolution, RT toggles, Lumen quality, shadows, textures, view distance, crowd density, HDR, FOV, motion blur, camera shake, film grain, gamma. Plus audio, controls, accessibility, language, a benchmark tool. All settings persist and are searchable.
- **Steam:** achievements, cloud saves, Steam Input, rich presence. Steam Deck compatibility as a stretch goal. Windows 10/11 64-bit, DirectX 12.
- **Dev tools:** console commands, cheat menu, free-fly cam, teleport, enemy spawner, performance HUD, traversal replay recorder.

17. OPTIMIZATION AND STREAMING

LODs, HLOD and imposters; occlusion culling; instancing; Nanite and VSM budgets; texture streaming pool budgets; PSO cache; async loading; time-sliced AI ticks; object pooling (VFX, projectiles, debris); audio voice limits; scalability groups mapped to hardware tiers. Profiling toolchain: Unreal Insights, RenderDoc, PIX / Nsight. Nightly performance captures on a fixed test route (fastest swing route, combat arena, crowd plaza) with regression alerts, plus 2-hour soak tests for memory leaks.

18. ACCESSIBILITY AND COMFORT

Follow the Game Accessibility Guidelines: all "basic" items from day one, "intermediate" where possible.

- **Controls:** remap every input; hold/toggle alternatives; sensitivity sliders; one-handed presets; simplified inputs; no-fail alternatives to timing-critical prompts.

- **Difficulty:** per-axis options (Section 6.2).
- **Audio/visual:** subtitles (size, background, speaker labels); separate volume controls; no information conveyed by audio alone or by color alone; colorblind modes; high-contrast UI; UI scaling; readable fonts.
- **Comfort:** FOV slider; reduce or disable camera shake and motion blur; horizon lock; steadier swing-camera option; photosensitivity warning and flash reduction.
- **Guidance:** skippable, re-openable tutorials; frequent checkpoints; screen-reader support for menus as a stretch goal.

19. LEGAL, IP AND RATING

- Spider-Man, Peter Parker, the supporting cast and the villains are Marvel/Sony property. Treat this as a private, learning or fan project unless properly licensed.

- Do not copy assets, stories, dialogue, music or code from existing games (including Insomniac's Spider-Man games) or films. Write original scripts, designs and audio. Keep an asset license tracker (source, license, proof).
- If I ever want to release or sell it: swap in an original web-slinging hero (new name, costume, power set, supporting cast) and keep the systems. Provide a swap checklist when I ask.
- Content rating: aim for ESRB T / PEGI 12. Stylized combat, no gore, no real-world brands.

20. PRODUCTION PLAN AND DEFINITION OF DONE

Do one phase at a time. Each phase ends with a playable build.

Phase 0: Pre-production (1 to 2 weeks).

Confirm engine, repo, folder structure, naming, coding standards, test route,

performance harness and risk list.

Done when: the project builds; a nightly build runs; the performance harness records a baseline.

Phase 1: Traversal prototype in a greybox district (4 to 6 weeks). Swing, zip, launch, wall-run, cling, glide (stub), camera, fast-travel stub.

Done when: 2 km of continuous swinging; 60 FPS on Baseline; feel $\geq 8/10$ from 5 testers; no traversal dead ends; live-tuning panel works.

Phase 2: Combat prototype (4 to 6 weeks). 5 enemy types, 1 mini-boss, combos, dodge / perfect dodge, web-strike, 4 gadgets, Spider-Sense, hit-feel.

Done when: a 1-versus-8 fight is readable and fun; attack tokens work; feel $\geq 8/10$; no performance regressions.

Phase 3: Vertical slice (8 to 12 weeks). One fully art-passed district; missions 1 to 4 including the Vulture boss; 3 activity types;

HUD and menus; save; settings; audio pass; final-quality lighting; a polished 30-minute demo.

Done when: 30 minutes at final quality; 60 FPS on Baseline; zero crashes or soft-locks; an external playtest completed.

Phase 4: Full production. Remaining districts, missions 5 to 16, all activities, suits and skills, cinematics, VO, music, events, progression balance. Monthly playable builds.

Done when: feature complete and content complete.

Phase 5: Online co-op (optional). See Section 15.

Done when: 2-player free-roam plus 3 missions are stable at 100 ms latency.

Phase 6: Polish and hardening. Optimization, QA, accessibility pass, localization, Steam integration.

Done when: performance gates are met on

all tiers; zero blocker or critical bugs; release-candidate build ready.

Phase 7: Release and live-ops. Patch pipeline; post-launch content (new district or villain arc); community feedback loop.

21. QA, TESTING AND TELEMETRY

- Automated tests: gameplay-math unit tests, functional tests, save/load tests, and traversal bots that swing test routes to find fall-throughs, streaming hitches and navigation gaps.
- Nightly smoke test; crash reporting; opt-in telemetry (death heat-maps, mission drop-off, performance by hardware).
- Bug severity policy: blocker / critical / major / minor.
- Playtests every 2 weeks with a 1 to 10 feel survey; observe, don't explain.

- Parity tests for controller and keyboard + mouse; a hardware matrix with at least one machine per tier.

22. OUTPUT FORMAT REQUIRED FROM YOU (THE AI)

For every deliverable provide: (1) goal and assumptions; (2) exact folder and file structure; (3) complete commented code or Blueprint setup with editor steps; (4) a tuning table with default values and what each does; (5) a test checklist with pass/fail criteria; (6) performance notes; (7) known limitations and the next step.

Use headings and short paragraphs. When I say "continue", move to the next item in the current phase. When I say "review", audit your last deliverable against Sections 3, 4.3 and 22 and flag anything that violates the pillars or budgets.

23. EXTRAS I DIDN'T ASK FOR BUT WANT (they make the game better)

Tutorial and first-10-minutes script · photo mode · per-axis difficulty · full accessibility suite · adaptive music · reactive radio/podcast · dynamic crime director · debug and cheat tools · benchmark tool · New Game+ · suit mods and cosmetics · time-trial ghosts · Steam achievements and cloud saves · localization-ready text · crash reporting · shader warm-up for a stutter-free start · predictive streaming · replay-based bug reproduction · content-rating checklist · asset license tracker · patch and update pipeline · post-launch content plan.

24. RISKS AND MITIGATIONS

- **Scope creep:** vertical slice first; freeze features per phase.

- **Traversal not fun:** tune before anything else; playtest weekly; keep the assists.
- **Open-world performance and stutter:** budgets as gates; predictive streaming; PSO cache; nightly profiling.
- **Content volume too large for a small team:** favor systemic activities; reuse through PCG and modular kits; cut side content before cutting story quality.
- **Online complexity:** offline first; co-op stays optional.
- **Asset quality and licensing:** license tracker; a consistent art bible.
- **IP exposure:** keep it private or swap in an original hero.

(End of Part A.)

PART B: PHASE PROMPTS (send one at a time after Part A)

Phase 0

"Begin Phase 0. Classify my PC (Section 0) into a hardware tier, pick and justify the engine and version, and give me: install steps, project creation settings, Git + LFS setup, folder structure, naming and coding standards, a performance test-route plan and a risk list. Then stop and wait for my go-ahead."

Phase 1

"Begin Phase 1: traversal prototype in a greybox city block. Build Section 6.1 completely: state machine, swing model with anchor selection, web-zip, point-launch, wall-run, wall-cling, ledge/vault, glide stub, camera with speed-scaled FOV, TraversalTuning data asset, live-tuning panel and debug HUD (speed, state, anchor). Give full code, editor steps, a tuning table and a test checklist. Show me how to measure the Definition of Done."

Phase 2

"Begin Phase 2: combat prototype. Build

Sections 6.2 to 6.4: attack chain, dodge and perfect dodge, parry, web-strike, aerial combat, Focus finishers, 5 enemy types with attack tokens, 1 mini-boss, 4 gadgets, Spider-Sense and hit-feel (hit-stop, camera, haptics, placeholder VFX/SFX). Give full code, setup steps, tuning table and tests."

Phase 3

"Begin Phase 3: vertical slice. One art-passed district, story missions 1 to 4 including the Vulture boss, 3 activity types, HUD and menus, save/load, settings plus benchmark, adaptive audio and final-quality lighting. Give the task breakdown first, then build in order of risk. Include an asset sourcing plan and a performance report against Section 4.3."

Phase 4

"Begin Phase 4: production. Produce a content plan (districts, missions 5 to 16, activities, suits, skill trees) with a spec per item, then implement in monthly milestones.

After each milestone give me a playtest script and a performance report."

Phase 5 (optional)

"Begin Phase 5: online co-op per Section 15. Explain the network model, build 2-player free-roam first, then missions. Give a latency and bandwidth test plan."

Phase 6

"Begin Phase 6: polish and hardening. Bring performance to the gates on all tiers, audit accessibility against Section 18, then handle localization, Steam integration, crash reporting and telemetry, and a release-candidate process."

Any time

"Review your last deliverable against my pillars (Section 3), performance gates (Section 4.3) and output format (Section 22). List violations and fix them."

PART C: GLOSSARY (so anyone reading this can follow)

- **Greybox:** a rough, untextured level made of simple shapes used to test gameplay before art.
- **Vertical slice:** one small section of the game built to final quality to prove the whole concept works.
- **Definition of Done (DoD):** the measurable checklist a phase must pass before moving on.
- **Nanite / Lumen / MegaLights:** Unreal Engine systems for detailed geometry, dynamic global lighting and reflections, and many dynamic shadowed lights.
- **PSO cache / shader stutter:** precompiled GPU state that prevents hitches when new effects first appear.

- **LOD / HLOD:** simpler versions of objects (or groups of objects) shown at a distance to save performance.
- **World Partition:** Unreal's system that streams a large world in and out in cells.
- **PCG:** procedural content generation; rules that place buildings, props and clutter automatically.
- **Motion Matching:** animation technique that picks the best-fitting clip pose every frame for smooth movement.
- **IK (inverse kinematics):** adjusts hands and feet to touch surfaces naturally.
- **Root motion:** animation that drives the character's actual movement.
- **Hit-stop:** a tiny freeze on impact that makes hits feel heavy.
- **Coyote time / input buffering:** forgiveness for slightly early or late button presses.
- **Attack tokens:** a limit on how many enemies may attack at once.

- **Spider-Sense:** the in-game danger sense that warns of attacks and enables dodge windows.
- **Perfect dodge / Focus meter:** a well-timed dodge that slows time; a meter that fills with good play and powers finishers.
- **Client-side prediction / reconciliation:** hiding network delay by moving instantly, then correcting gently to the server's version.
- **Listen server:** one player's PC hosts the co-op game.
- **Iris:** Unreal's newer network replication system.
- **Upscaling (DLSS / FSR / XeSS):** rendering at lower resolution and reconstructing a sharper image.
- **Frame generation:** inserting extra AI-made frames to raise displayed FPS (adds latency, so optional).
- **1% lows / frame time:** the worst-case smoothness measure; the milliseconds each frame takes.

- **Telemetry:** anonymous usage and performance data (opt-in).
 - **NG+:** New Game Plus, a replay with earned upgrades.
 - **Data asset:** a config file of tunable numbers editable without code.
 - **Behavior Tree / StateTree / EQS:** tools for enemy decision-making and environment queries.
-

PART D: RESEARCH NOTES AND SOURCES (checked 28 Sep 2026)

- **Engine status:** UE 5.8 (June 2026) is the last planned major UE5 release; UE6 early access targeted for end of 2027. 5.8 highlights: MegaLights production-ready, Lumen Lite, Iris production-ready, Mover improving, MetaHuman crowds (experimental), experimental MCP plugin.

- unrealengine.com/news/state-of-unreal-2026-top-news-from-the-show
- unrealengine.com/news/unreal-engine-5-8-is-now-available
- dev.epicgames.com/documentation/unreal-engine/unreal-engine-5-8-release-notes
- tomlooman.com/unreal-engine-5-8-performance-highlights/
- **Unity:** 6.3 LTS supported until December 2027. unity.com/releases/unity-6/support
- **PC benchmark (Marvel's Spider-Man 2 PC):** support.nixxes.com (system requirements); pcgamebenchmark.com/marvel-s-spider-man-2-system-requirements. PC features include DLSS 3, FSR 3.1, XeSS, ray-traced reflections/shadows/AO, ultrawide up to 48:9 and dynamic resolution scaling.
- **Real-world PC hardware:** Steam Hardware Survey, July 2026 (RTX 3060 most popular GPU; 8-core CPUs and 16 GB-

VRAM GPUs now the most common configurations) and June 2026 (16 GB system RAM most common).

- **Traversal design lessons (GDC):**
"Concrete Jungle Gym: Building Traversal in Marvel's Spider-Man"; "Classic Game Design Postmortem: Swinging with Spider-Man" (swinging tends to be too hard and breaks flow); "Higher, Faster, Farther: Evolving Traversal in Marvel's Spider-Man 2" (web wings and adapting systems to higher speeds). gdcvault.com and schedule.gdconf.com.
- **Accessibility:**
gameaccessibilityguidelines.com